Lucie Choi

luciechoi.xyz | **clucie0407@gmail.com**

Education

University of California, Berkeley — Berkeley, CA
BA, Computer Science (Focus in Systems) | *GPA: 3.65* — *2016 – 2020*

Stanford University — Stanford, CA
Graduate Non-Degree Option (NDO), Computer Science *(Focus in Visual Computing)* | *GPA: 3.67* — *2025 – 2026*

- **Coursework:** CS 348C (Advanced Animation and Simulation), CS 448Z (Physically Based Sound), CS 248B (Character Animation and Simulation), CS 248A (Rendering and Geometry), CS 148 (Intro Computer Graphics), EE 263 (Matrix Methods)

Professional Experience

Google — San Jose, CA
Senior Software Engineer, Android XR Graphics — *Dec 2021 – Present*

- **XR Glasses Graphics (2026 – Present):** Architecting low-latency, power-efficient Android graphics and display pipelines in XR glasses. Integrating Qualcomm hardware-accelerated Late Stage Reprojection (LSR) to minimize motion-to-photon latency.
- **Shader Compilers & Vulkan Backend (2025–2026):** Contributed to open-source **DXC** and **LLVM** compilers for HLSL-to-SPIR-V translation, implementing `CombinedImageSampler` support and GPU reconvergence features for Vulkan.
- **Likeness Photorealistic Avatars (Tech Lead, 2024–2025):** Led end-to-end backend and mobile app development for avatar creation and data streaming between 1st party VR headsets and Android phones. **Press Coverage**
- **YouTube VR (Tech Lead, 2023–2024):** Engineered real-time video co-watching engine for Meta Quest headsets, including playback synchronization, party management, and data encryption. **Press Coverage**
- **Document AI Core Infrastructure (2021–2023):** Developed backend services to allow enterprise customers to manage domain-specific documents and run various machine learning tasks, such as OCR, parsing, and classification.

Intuit — Mountain View, CA
Software Engineer, Full-Stack — *Jun 2020 – Oct 2021*

- Designed backend microservices and data migration to securely transition customers' banking data acquisition methods.

UC Berkeley RISELab — Berkeley, CA
Undergraduate Research Assistant, Database Systems — *Jan 2019 – Dec 2019*

- Co-developed **DIEL**, a declarative framework defining UI interactions as event tables via SQL. Built a query compiler and runtime that supports dynamic view materialization across multiple client (SQLite/sql.js) and server (PostgreSQL) backends.

Selected Projects

Cloth Simulation: Position-Based Dynamics — Stanford CS 348C

- Implemented a custom Position-Based Dynamics (PBD) cloth solver from scratch in Python featuring stretch, bend, and shear constraints, spatial hashing for self-collisions, and object collision, rendered in Houdini.

Temple Bell Sound Simulation: Modal Sound Synthesis & Acoustic Transfer — Stanford CS 448Z

- Simulated modal sound and acoustic transfer of the "Bell of King Seongdeok" using Finite Element Method (FEM) and Equivalent Source Method (ESM). Created a full audio-driven particle visualization in SciPy and Houdini.

Selected Skills

Graphics APIs & Systems: Vulkan, OpenGL, Android Graphics (AOSP), Taichi Lang, ThreeJS
Digital Content Creation (DCC) Tools: Houdini, Maya, Davinci Resolve
Languages: C++, Python

Leadership & Community

Google Pride Korea — Remote / Seoul, South Korea
Committee Member — *2026 – Present*

- Organize the company's various LGBTQ+ initiatives in South Korea—including Seoul Queer Culture Festival, Pride Parade Walk, and in-company events—championing inclusion within Google while inspiring greater diversity across Korean society at large.